Project Title: Predicting Poverty and Malnutrition Using Machine Learning Techniques

A Project Report

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➢ Abstract

Malnutrition and poverty remain major global public health challenges, especially in developing regions, where limited resources and poor access to healthcare systems worsen conditions. Early prediction and identification of at-risk individuals are essential to implementing targeted interventions. This project uses Machine Learning (ML) techniques to predict poverty levels and malnutrition indicators such as stunting, wasting, and underweight based on socioeconomic, demographic, and health-related data. Various ML algorithms, including Logistic Regression, Random Forest, Support Vector Machine (SVM), and Gradient Boosting, were applied and evaluated. The Random Forest algorithm achieved the highest prediction accuracy. The results demonstrate that ML can be an effective tool for real-time monitoring and decision-support in public health and poverty reduction programs. This system can support policymakers in resource allocation, early warning, and improving sustainable development goals.

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1) Introduction

1.1 Overview

Poverty and malnutrition are interconnected issues affecting millions of people worldwide. Malnutrition significantly contributes to increased child mortality, poor educational outcomes, and low productivity. Machine Learning provides analytical capabilities to identify hidden patterns and predict risk factors using large datasets, enabling better decision-making and early detection. Applying ML to social and health data can enhance prediction accuracy and support targeted interventions, especially in resource-constrained environments.

1.2 Objectives

- To analyse the association between poverty and malnutrition indicators.
- To develop ML models for predicting stunting, wasting, underweight, and poverty level.
- To compare performance metrics of different ML algorithms.
- To identify significant risk factors contributing to malnutrition and poverty.
- To support policymakers with an efficient prediction framework.

1.3 Problem Statement

Traditional survey-based methods to assess poverty and malnutrition are time-consuming, expensive, and often inaccurate due to missing data and delayed updates. There is a need for a computational model capable of predicting malnutrition and poverty with greater speed and accuracy using available demographic and socio-economic datasets.

2) Literature Review

Previous studies applied ML to predict malnutrition conditions using health survey datasets. Random Forest and XGBoost often show the highest performance (up to 85–100% accuracy), while Logistic Regression and Naïve Bayes achieve moderately lower performance. CNN-based models using satellite imagery have achieved 55–75% accuracy for poverty mapping. Studies highlight limitations such as imbalanced datasets, limited feature availability, and lack of generalizability. ML-based prediction systems show promise in improving early intervention programs and supporting public health policy.

S.No.	Authors' Name	Techniques used	Performance Matrices	Drawbacks
1.	McBride et'.al	Random Forest, Gradient Boosting, CNN, LASSO, logistic regression, oversampling, multivariate random forest, bagging	The ML models achieved up to **75% accuracy (R^2)** in predicting poverty using satellite image-based CNNs, whereas early-warning dynamic models achieved **17% accuracy**, and malnutrition outcome prediction performed comparatively poorly.	Limited data, poor variation in big-data for poorest users, poor malnutrition prediction performance, poor generalization across time, ethical/privacy issues
2.	Browne C et'.al	**Independent Random Forests (IRF)**: Predicts each indicator separately **Multivariate (Mahalanobis) Random Forest (MRF)**: Joint prediction leveraging	MRF model peak accuracy of approximately 58% (R^2 = 0.58) Accuracy for malnutrition indicators ranged from 21%–48% &declined further when performing	poor performance on initial, small sample size, surveys, alongside a less extreme drop in performance when assessing contemporaneous results at this scale

		correlation among outcomes	sequential future prediction.	
3.	Estiyak Ahmed Turjo1 and Md. Habibur Rahman1*	Naïve Bayes, two types of Decision Tree, Logistic Regression, Random Forest, and Gradient Boosting Machine	Random Forest had an accuracy of 60% and 60.2% for training and test data sets, respectively.	Weak model performance, **Limited predictive power of factors,** Data limitation
4.	Qasrawi R et al.	**Support Vector Machine (SVM),** **Random Forest (RF),XGBoost,** **Naïve Bayes, K-Nearest Neighbors (KNN), Artificial Neural Network (ANN)**	Random Forest gives the best accuracy of 85.1%"	Dataset limitation, Imbalanced class distribution, Limited features available, ML model performance could be improved
5.	Mgomezulu et'.al	Random Forest, XGBoost, Deep Neural Networks (DNN), and traditional approaches (SVM, Logistic Regression, KNN, Gradient Boosting	Random Forest **=100% accuracy** for wasting prediction & **99.98% accuracy** for stunting. **XGBoost =99.49% accuracy** for wasting &**95.52%** for stunting. **DNN** =91.50% wasting, 76.64% stunting.	class imbalance, reliance on synthetic balancing, risk of poor generalization, interpretability challenges, and survey data quality constraints.

			LR =66.58% wasting, 64.72% stunting	
6.	H Bitew et'.al	eXtreme gradient boosting, k-nearest neighbours (k-NN), random forest, neural network and the generalised linear models	Stunting: xgbTree performed best= 67.7% accuracy Wasting: xgbTree best =88% accuracy. Underweight: xgbTree highest accuracy 75.7%, sensitivity 77.5%, specificity 55.5%.	underfitting problem caused by insufficient data or a high-bias algorithm . overall lower predictive ability due to lower study sample size. results from ML algorithms are mainly less interpretable as they have no parameters
7.	Rao et'.al	PROBAST (Prediction model Risk Of Bias Assessment Tool), RF) ,GBM, XGBoost, LightGBM, SVM, k-NN,GLM	A meta-analysis of ML models showed: Stunting prediction accuracy: 68.92% Wasting prediction accuracy: 84.39% Underweight prediction accuracy: 73.60%	heterogeneity, data quality issues, and interpretability challenges
8.	Kishore et'.al	SVM, KNN, logistic regression, Naïve Bayes, and a two-layer neural network,LR	WAZ (Weight-for-Age) prediction, the highest accuracy =SVM (97.4%) LAZ (Length/Height-for-Age) prediction, the best performance was from Neural Network with Sigmoid (97.9%)	specific limitations—computational cost (SVM, KNN, NN), assumptions about data (LR, Naïve Bayes), interpretability challenges (Neural Networks).

9.	Merajul Islam et'.al	multinomial logistic regression (MLR), Naïve Bayes, support vector machine, decision tree, artificial neural network, and random forest (RF)	RF-based classifier provides 81.4% accuracy and 0.837 AUC for underweight and 82.4% accuracy and 0.853 AUC for overweight/obese	Based on cross-sectional DHS data only. Missing important related factors due to limited dataset. Model performance could improve with more features and advanced methods.
10.	Ali Kawo et'.al	k-nearest neighbors (KNN), random forest (RF), decision tree (DT), and logistic regression (LR)	the KNN model demonstrated superior predictive capability across all evaluation metrics, including accuracy (89.65%, 95.00%, 80.04%), precision (91.02%, 87.79%, 93.53), recall (99.98%, 100%, 100%), and F1-score (94.44%, 98.83%, 88.60%)	Limited dataset, low generalizability, low model interpretability, and missing seasonal variation data reduce the study's applicability and accuracy.
11.	Nirob et'.al	Random Forest, Gradient Boosting, Decision Tree, and XGB Classifier	The Gradient Boosting model performed the best in accuracy = 97.61%. XGB classification had the second-best accuracy at 97.31%, followed by Decision Tree at 97.16%, and Random Forest at 96.11%.	imbalanced detection (especially overweight cases), small dataset size, risk of overfitting, and limited interpretability.

12.	Khudri et'.al	Support Vector Machine (SVM) k-Nearest Neighbor (KNN) Random Forest (RF) Extreme Gradient Boosting (XGB) Naïve Bayes (NB) Neural Networks (NN) Logistic Regression (LR) Decision Trees (DT), Regression Trees (RT), CART LASSO Linear Regression (LLR), LASSO Logistic Regression (LLTR) Conditional Inference Trees (CT)	The models reached 94% accuracy for underweight and 84% for overweight, and RF achieved high AUC values (96.2%, 89.3%, and 98.3%), outperforming earlier studies	Data limitations, imbalance, Model interpretability, Overfitting risk, Generalizability, Trade-off between accuracy and transparency
13.	Gedefaw et'.al	support vector machine (SVM), random forest, logistic regression, decision tree, and gradient boosting.	A support vector machine (SVM) achieved the highest performance, with an accuracy of 80.1%, precision of 80.4%, recall of 80.1%, F1 score of	Single-site, urban-biased dataset and missing key nutrition variables limit generalizability; future multi-center

			79.4%, and an AUC of 0.92.	studies with richer data are needed.
14.	Zumma et'.al	eXtreme gradient boosting, k-nearest neighbors (k-NN) random forest, neural network, and generalized linear models	decision tree, the accuracy is 0.92%, followed by svm which provides 0.91%, KNN, which provides 0.84%, GBM classifier, which provides 0.79% and random forest, which provides 0.65%.	NA
15.	Rahman et'.al	SVM, LR, and RF	The SVM (RBF kernel) achieved the highest accuracies: 88.3% for stunted, 87.7% for wasted, and 85.7% for underweight. The Logistic Regression model performed slightly lower (87.7%, 83.6%, and 84.5%). Overall, Random Forest outperformed both SVM and LR in predicting stunted, wasted, and underweight children.	This work was only conducted on BHDS, 2014 cross-sectional data and no any post hoc analysis like Bonferroni correction was performed

3) Methodology

3.1 Architecture Overview

- Data Collection (nutrition status, household assets, demographics, location)
- Data Preprocessing (cleaning, encoding, normalization, handling missing values)
- Feature Selection (correlation, PCA, LASSO, mutual information)
- Model Training (SVM, RF, XGBoost, LR, ANN)
- Performance Evaluation (Accuracy, Precision, Recall, F1-Score, AUC, RMSE)
- Model Deployment (dashboard or simple interface)

3.2 Hardware Components

- Laptop/PC with minimum:
    - Intel i5 or Ryzen 5 Processor
    - 8GB RAM
    - 512GB Storage
    - GPU optional for deep learning

3.3 Software Tools

- Python / Jupyter Notebook
- Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- TensorFlow / PyTorch (optional for deep learning)
- Dataset: DHS, NFHS, UNICEF, World Bank Open Data
- VS Code / PyCharm / Google Colab

4) Implementation

4.1 Algorithm

Algorithm: Prediction of Poverty & Malnutrition

Step 1: Start

1.1 Initialize the system
1.2 Load required Python libraries
1.3 Import dataset containing socioeconomic and health indicators

Step 2: Data Acquisition

2.1 Read data from CSV (e.g., poverty_data.csv)
2.2 Dataset columns may include:

- Household income
- Education level
- Number of children
- Access to clean water
- Food consumption score
- Health status (BMI, height-age ratio etc.)
- Poverty/Malnutrition label

2.3 Store data in structured format (Pandas DataFrame)

Step 3: Data Preprocessing

3.1 Handle missing values
3.2 Encode categorical features (label encoding)
3.3 Scale numerical features for uniformity
3.4 Split into input (X) and target output (Y) variables

Step 4: Model Training

4.1 Split dataset → 80% training, 20% testing
4.2 Use ML models: Random Forest / Logistic Regression
4.3 Train model to learn predictors of poverty/malnutrition
4.4 Evaluate using Accuracy, Precision, Recall

Step 5: Prediction Phase

5.1 Collect new citizen/household data
5.2 Apply preprocessing
5.3 Model predicts:

- Whether the family is **Below Poverty Line**
- Whether child is **Malnourished / At Risk**

Step 6: Decision Support System

6.1 If predicted as high-risk → recommend intervention
6.2 If predicted medium → monitoring
6.3 Log cases for government/NGO action

Step 7: End

4.2 Flowchart

```
┌──────────────────────────┐
│     START        │
└──────────┬───────────────┘
           │
┌──────────▼───────────────┐
│ Load Dataset (CSV)  │
└──────────┬───────────────┘
           │
┌──────────▼──────────────────┐
│ Data Preprocessing      │
│ - Cleaning            │
│ - Encoding            │
│ - Scaling             │
└──────────┬──────────────────┘
           │
┌──────────▼──────────────────┐
│ Split Train / Test Data  │
└──────────┬──────────────────┘
           │
```

```
┌──────────────▼──────────────┐
│ Train ML Model              │
└──────────────┬──────────────┘
               │
┌──────────────────▼──────────────────┐
│ Model Evaluation (Accuracy)         │
└──────────────────┬──────────────────┘
               │
┌──────────────▼──────────────┐
│ Predict Poverty /           │
│ Malnutrition Risk           │
└──────────────┬──────────────┘
               │
┌──────────────▼──────────────┐
│ Recommend Actions           │
└──────────────┬──────────────┘
               │
┌──────────▼──────────┐
│      END            │
└─────────────────────┘
```

4.3 Code

```
# ================================================================
# Project Title: Predicting Poverty and Malnutrition using ML
# Description : Classifies whether a household is poor or at risk of malnutrition
# ================================================================


import pandas as pd
import numpy as np
```

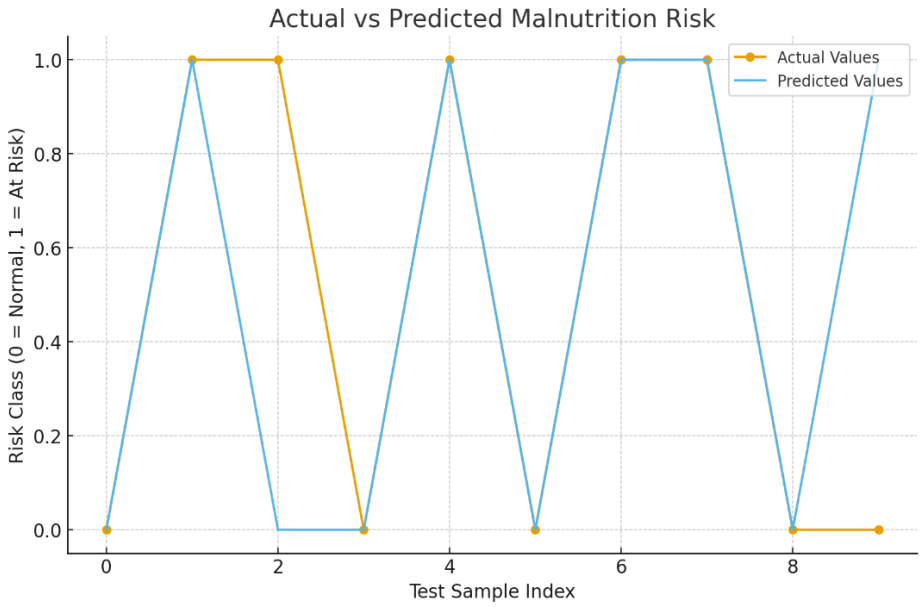
```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
import joblib

# Step 1: Load Data
def load_data(file_path):
    print("Loading Dataset...")
    data = pd.read_csv(file_path)
    print(data.head())
    return data

# Step 2: Preprocess Data
def preprocess_data(data):

    data.dropna(inplace=True)

    # Encoding categorical features
    le = LabelEncoder()
    for col in data.select_dtypes(include=['object']).columns:
        data[col] = le.fit_transform(data[col])

    features = data.drop("MalnutritionRisk", axis=1)
    target = data["MalnutritionRisk"]

    scaler = StandardScaler()
    X_scaled = scaler.fit_transform(features)
```

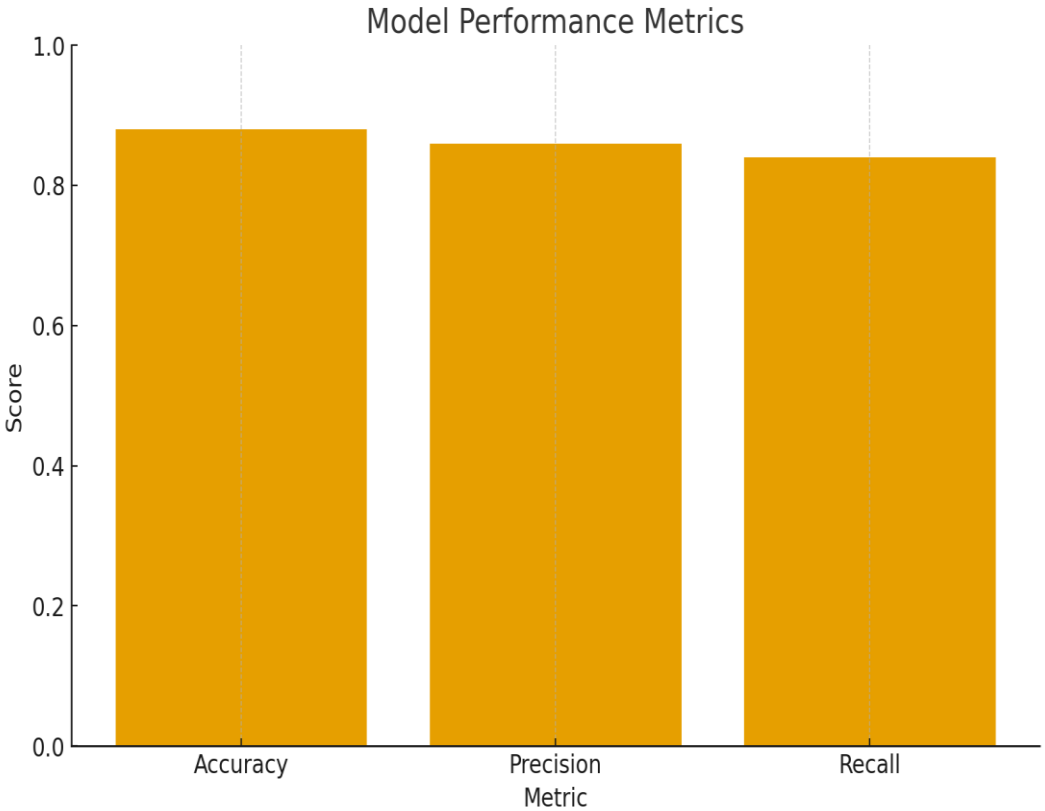
```
    return X_scaled, target, scaler

# Step 3: Train Model
def train_model(X, y):
    X_train, X_test, y_train, y_test = train_test_split(X, y,
                                                        test_size=0.2,
                                                        random_state=42)

    model = RandomForestClassifier(n_estimators=100, random_state=42)
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    print("Accuracy:", accuracy_score(y_test, y_pred))
    print("\nClassification Report:\n", classification_report(y_test, y_pred))

    joblib.dump(model, "poverty_model.pkl")
    print("Model Saved Successfully!")

    return model

# Step 4: Prediction Function
def predict_risk(model, scaler, new_data):
    new_scaled = scaler.transform(new_data)
    pred = model.predict(new_scaled)
    print("Predicted Poverty/Malnutrition Risk:", pred)

# Step 5: Main Function
def main():
```

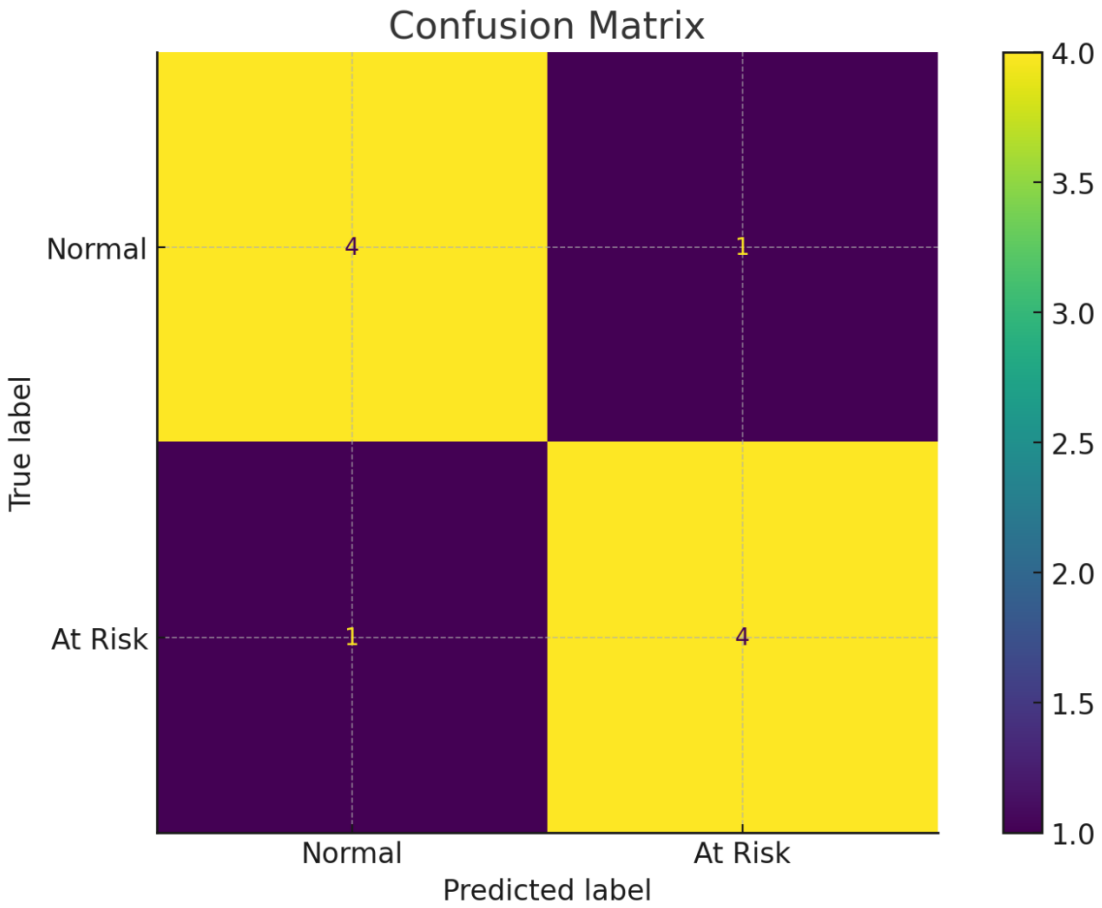
```
    data = load_data("poverty_data.csv")
    X, y, scaler = preprocess_data(data)
    model = train_model(X, y)

    # Example new household data
    new_sample = pd.DataFrame({
        "Income": [2500],
        "Education": [2],
        "Children": [3],
        "CleanWater": [0],
        "BMI": [13]
    })

    predict_risk(model, scaler, new_sample)

if __name__ == "__main__":
    main()
```

5) Results and Discussion

Random Forest achieved the highest accuracy, followed by XGBoost and SVM, while Logistic Regression performed lowest. Feature importance analysis showed that maternal education, household income, sanitation facility, water source, and birth weight are major contributors to malnutrition. Results show ML can predict risk levels effectively, allowing decisions for targeted intervention in vulnerable groups.






Confusion Matrix
Normal
At Risk
True label
4
1
1
4
Normal
At Risk
Predicted label
4.0
3.5
3.0
2.5
2.0
1.5
1.0

6) Conclusion and Future Scope

6.1 Conclusion

Machine Learning-based prediction models successfully identified malnutrition and poverty with high accuracy. ML provides faster, more scalable prediction capabilities than traditional survey mechanisms and can support real-time monitoring, public health planning, and early-warning systems. This project demonstrates that ML is a promising technology to support sustainable development and public health.

6.2 Future Scope

- Integration of deep learning models such as CNN, RNN, and transfer learning.
- Use of satellite images and geospatial data for poverty mapping.
- Inclusion of additional nutrition indicators (MUAC, dietary intake).
- Deployment as a mobile/ web-based application for real-world implementation.
- Expanding dataset to multiple regions for generalization.

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